

MATHEMATICS FOR AI/ML

Second Edition

This comprehensive, full-color textbook provides a rigorous mathematical foundation for modern artificial intelligence and machine learning. Covering topics from linear algebra and optimization to stochastic differential equations and generative models, it bridges mathematical theory with practical applications in deep learning, reinforcement learning, large language models (LLM), and generative AI.

Key Features

- Graduate-level coverage of linear algebra, optimization, matrix calculus and information theory
- Advanced topics including SDEs, Fokker-Planck equations, and score-based generative models
- Colorful figures with AI/ML examples throughout
- Extensive exercises and problem sets for self-study and instruction
- Mathematical rigor with practical insights for AI/ML research and practice

Intended Audience

Designed for graduate students, AI/ML researchers, and experienced practitioners seeking a rigorous mathematical foundation. The book also supports Olympiad-style mathematical training and can serve as a supplementary reference for advanced AI competition preparation at the national and international level.

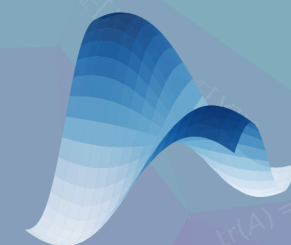
MATHEMATICS FOR AI/ML

Henry Fuheng Wu

MATHEMATICS FOR AI/ML

Second Edition

A Graduate-Level, Full-Color Reference
for Machine Learning and Artificial Intelligence



By Henry Fuheng Wu

2026